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SUPERSEDING
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May 28, 1964

FEDERAL SPECIFICATION

BLACKBOARDS, PORTABLE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers portable blackboards with writing surfaces of composition and porcelain enameled steel both stand-mounted and wall-mounted (see 6.1).

1.2 Classification.

1.2.1 Types, styles and sizes. The blackboards shall be of the following types, styles and sizes, as specified (see 6.2).

Type I - Porcelain enameled writing surface, with magnetic holders.
  Style A - Wood stand-mounted.
  Style B - Aluminum stand-mounted.
    Size 1 - 36 inches high by 48 inches long.
    Size 2 - 48 inches high by 72 inches long

Type II - Composition writing surface
  Style A - Wood stand-mounted.
  Style B - Aluminum stand-mounted.
    Size 2 - 48 inches high by 72 inches long.
  Style C - Wall-mounted, wood frame
    Size 2 - 48 inches high by 72 inches long
    Size 3 - 48 inches high by 120 inches long.
  Style D - Wall-mounted, aluminum frame.
    Size 2 - 48 inches high by 72 inches long

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications

UU-P-268 - Paper, Kraft, Untreated, Wrapping.
LLL-B-810 - Building Board, (Hardboard) Hard Pressed, Vegetable Fiber.
PPP-B-591 - Boxes, Shipping, Fiberboard, Wood-Cleated.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-C-650 - Crates, Wood, Open and Covered.
PPP-F-320 - Fiberboard· Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.
PPP-P-291 - Paperboard, Wrapping and Cushioning.
PPP-T-76 - Tape, Pressure-Sensitive Adhesive Paper, (for Carton Sealing).

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

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(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies).

Military Specifications.

MIL-P-116 - Preservation, Methods of.
MIL-L-10547 - Liners, Case and Sheet, Overwrap, Water-Vaporproof, or Waterproof, Flexible

Military Standards

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-130 - Identification Marking of U S. Military Property.
MIL-STD-1363 - Measurement of Wood Moisture Content.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

United States Department of Commerce

Commercial Standard CS45-60 - Douglas Fir Plywood

(Application for copies should be addressed to the Superintendent of Documents, U S. Government Printing Office, Washington, DC 20402)

Porcelain Enamel Institute

PEI S-100 - Architectural Porcelain Enamel on Steel for Exterior Use.
Bulletin T-16 - Gloss Test for Porcelain Enamel

(Application for copies should be addressed to the Porcelain Enamel Institute, Inc , 1900 L. Street, N.W , Washington, DC 20036.)

National Motor Freight Traffic Association, Incorporated, Agent

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc , Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified (see 6.2), before production is commenced, a sample of each type and style of the finished commodity on contract and each component part shall be submitted or made ready for the contracting officer or his authorized representative to examine and test to determine compliance with this specification. Approval of the preproduction sample authorizes the commencement of production, but does not relieve the contractor of responsibility for compliance with all other applicable provisions of this specification. Production units shall not vary from the approved preproduction sample in design or construction without written approval of the contracting officer.

3.2 Material. All materials used shall be as hereinafter specified. All materials shall be unused and free from defects which affect serviceability or appearance of the finished product.

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3.2.1 Wood. Wood shall be oak, beech, birch, hackberry, hard maple, pecan, or ramin. The wood shall be kiln-dried to a moisture content of 5 to 9-percent at time of assembly and shall be without brashness, case-hardening, or honey-combing. Moisture content shall be determined in accordance with MIL-STD-1363. Wood shall be plain or quarter sawed. Slope or divergence of grain shall not exceed 1 inch in 5 inches in any one surface of the unit when tested in accordance with 4.5.1. The wood components shall be free from any defects except that the following will be permitted:

(a) Sound, tight knots not exceeding 1/4-inch in greatest dimension and not less than 5 inches apart on any face.
(b) Sap strain or discoloration, provided that not more than two such defects appear in any one component part.
(c) Not more than two wormholes per component, 1/16-inch or less in diameter, provided they do not affect serviceability.
(d) Small pin or twig knots, not exceeding 3/32-inch diameter. These imperfections shall not cause cracks nor grain separations in the wood. Groups or clusters of these faults in the wood are permitted but the individual faults shall not be closer to one another than 1/4-inch.

3.2.2 Plywood. Plywood for filler shall conform to requirements for interior type, grade AA, of Commercial Standard CS45-60.

3.2.3 Hardboard. The hardboard shall conform to type I of LLL-B-810.

3.2.4 Porcelain enamel blackboard. The porcelain enamel coating shall be in accordance with Porcelain Enamel Institute, Inc., PEI: S-100, except that packing and guarantee requirements specified therein shall not apply.

3.2.5 Hardware

3.2 5.1 Casters. Casters shall be ball-bearing, non-marking rubber-wheeled, swivel type. At least two of the casters on each blackboard shall have a locking device to prevent movement of the unit.

3.2 5.2 Glides Glides shall be nail-on type The lower face shall be stainless steel, nylon, molded phenolic plastic or high impact styrene plastic The stem shall be serrated and 7/8-inch long, minimum The base diameter shall be not less than 1-1/8-inch.

3 2.5.3 Magnetic holders. The magnetic holders shall measure 1 X 3/8 X 7/32-inch thick ± 1/16-inch and shall have a pull of approximately 10 ounces.

3.3 Construction.

3.3.1 Design The design and dimensions of the blackboards shall be as shown in attached drawings and as specified herein. The surface texture of the board shall permit ease of writing, take chalk evenly, erase easily and be resistant to wear and chipping in normal use The blackboards shall be ready for use without any use preparations.

3.3.1.1 Type I, Porcelain enamel writing surface, with magnetic holders The design and dimensions of the blackboards shall be as shown in figure 1 and figure 2. The blackboards shall be reversible and shall have a writing surface on each side. The writing surface shall be as specified in 3.2.4 and the color of the finish shall be medium green. The porcelain shall be fused to one side each of two sheets of 24-gage enameling steel. The uncoated surfaces of the two steel sheets shall be firmly and completely bonded to a filler of not less than 1/4-inch plywood specified in 3.2.2, or a filler of hardboard specified in 3.2.3. The bonding material used shall provide permanent adhesion. The standards and frames shall be in accordance with 3.3.4 and 3.3.6. The standards shall be equipped with casters specified in 3.2.5.1 and shall be furnished with not less than 12 magnetic holders specified in 3.2.5.3.

3.3.1.2 Type II, Styles A and B blackboards. The design and dimensions of the blackboards shall be as shown in figure 1 and figure 2. The blackboards shall be reversible and shall have a writing surface on each side The writing surface shall be a composition coating consisting of such material as ground slate, carborundum or equivalent material and a binder. The gloss of the blackboards shall not increase by more than three units when tested as specified in 4.5.2. The color of the finish of the writing surface shall be medium green or black as specified (see 6.2). The composition coating shall be applied to both face surfaces of a single piece of surface 2 (smooth two sides) hardboard of the type specified in 3.2.3. The hardboard shall be 1/4-inch thick, or alternative, two pieces of 1/8-inch thick hardboard securely bonded together may have the composition applied to both outer surfaces. The standards and frames shall be in accordance with 3.3.4 and 3.3.6. The standards shall be equipped with casters specified in 3.2.5.1, or with glides specified in 3.2.5.2, as specified in the invitation for bids (see 6.2).

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3.3.1.3 Type II, Style C blackboard. The blackboard shall have a single writing surface which shall be a composition coating consisting of such material as ground slate, carborundum or equivalent material and a binder. The gloss of the blackboard shall not increase by more than three units when tested as specified in 4.5.2. The color of the finish of the writing surface shall be black. The composition coating shall be applied to the face surface of hardboard specified in 3.2 3. The 48 by 72-inch size shall be 1/4-inch thick hardboard or two pieces of 1/8-inch thick hardboard securely bonded together. The 48 by 120-inch size shall be 1/2-inch thick hardboard obtained by securely bonding two pieces of 1/4-inch thick hardboard. The wood frame shall be in accordance with 3 3 5 and chalk trough shall be in accordance with 3 3.5.1.

3.3.1.4 Type II, Style D blackboard. The blackboard shall have a single writing surface which shall be a composition coating consisting of such material as ground slate, carborundum or equivalent material and a binder The gloss of the blackboard shall not increase by more than three units when tested as specified in 4.5.2. The color of the finish of the writing surface shall be green The composition coating shall be applied to the face surface of a single piece of hardboard specified in 3.2.3. The hardboard shall be 1/4-inch thick hardboard or two pieces of 1/8-inch thick hardboard securely bonded together. The aluminum frame shall be in accordance with 3.3.6 3 and the chalk trough shall be in accordance with 3.3.6.4.

3 3.2 Dimensions, Type II blackboards. The style A and style B blackboards shall be size 2, 48 inches high by 72 inches long ± 1 inch. The style C blackboard shall be size 2, 48 inches high by 72 inches long ± 1 inch, and the size 3 blackboard shall be 48 inches high by 120 inches long ± 1 inch. Style D blackboards shall be size 2, 48 inches high by 72 inches long, ± 1 inch

3 3 3 Tolerances. A tolerance of ± 1/16-inch is permitted on width and thickness dimensions of wood members and cross-section of aluminum members.

3 3.4 Wood standard and frame

3 3 4.1 Standard. The standard shall be constructed of wood conforming to 3 2.1 The vertical uprights shall be 2 inches wide by 15/16-inch thick Each upright shall be supported by a base The base shall be of solid or built-up stock and shall be 2 inches high and 1-1/2 inches wide The ends shall be chamfered or rounded The vertical uprights shall be mortised and tenoned or double-doweled and glued to the base The base and upright assembly shall be provided with additional support in the form of braces secured to the edge of the upright and the upper face of the base The braces shall be attached at each end by glue and two dowels or glue and two screws There shall be a horizontal brace, incorporating a chalk trough, connecting the two vertical uprights The top of the chalk trough shall be 30 inches from the floor. The bottom edge of the blackboard frame shall be 3 inches above the chalk trough . The horizontal brace shall be shaped to provide a chalk trough accessible from both sides of the blackboard and shall run the full width. The trough shall be of sufficient size to hold chalk and standard size erasers, exclusive of groove The trough shall have an inside groove to allow the accumulation of chalk dust below the level at which the eraser rests The horizontal brace shall be attached to vertical uprights by means of 1/4-inch bolts and nuts, screws, glue and dowels, or any other means that will provide for the permanent secure attachment of the two members Support shall be provided between each vertical upright and horizontal brace by means of two braces Each brace shall be attached to the horizontal brace and the vertical uprights Any means of attachment may be used in securing the brace such as bolts and nuts, screws, dowels and glue, provided the attachment is secure and produces a stable, strong unit All mechanical fastners, when used must be countersunk and flush with the surface of the material Full height vertical uprights shall have a cross-tie above the top of the blackboard frame which shall be 2 inches wide by 15/16-inch thick. The cross-tie shall be securely attached Alternatively, the top cross-tie may be eliminated when less than full height vertical uprights are used All exposed edges of the standard shall be rounded or chamfered a minimum of 1/6-inch

3.3 4.2 Wood frame The blackboard frame shall be constructed of wood conforming to 3.2.1 The frame shall totally enclose the blackboard writing surface The frame shall be a minimum of 1-3/4 inches wide by 3/4-inch thick The inner edges of the frame shall be dadoed to receive the blackboard panel The blackboard shall be secured to the frame by a method which secures the blackboard around its entire perimeter and prevents any movement of the blackboard The attachment method shall not weaken the frame in any manner. The corners of the frame may be mitered and joined by spline or dowels, or the corners may be joined by open or blind mortise and tenon. All exposed edges shall be rounded a minimum of 1/8-inch When the pivoting device used is designed so that the threaded stem must penetrate the vertical frame member, each vertical frame member shall have an embedded metal-threaded insert to receive the stem

3.3.4.3 Pivoting device and locking device. Each vertical upright of the standard shall be equipped with an easily insertable 1/4-inch diameter, key-type threaded pivot-pin which will permit the frame and blackboard to pivot on its horizontal axis a full 360 degrees. The pivot-pin assembly shall incorporate a locking device of a positive type which will hold the blackboard securely in a vertical position. Alternatively, the blackboard shall be secured in a vertical position by means of a permanently attached spring-loaded steel pin, or other mechanical device, which in the engaged position shall lock blackboard in a vertical position and which will permit disengaging and movement of blackboard until manually re-

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3 3.5 Wall-mounted wood frame. The blackboard frame shall be constructed of wood conforming to 3.2.1. The frame shall totally enclose the blackboard writing surface. The frame shall be a minimum 1-3/4 inches wide by 3/4 inch thick The inner edges of the frame shall be rabbeted or dadoed to receive the blackboard panel The blackboard shall be secured to the frame by a method which secures the blackboard around its entire perimeter and prevents any movement of the blackboard. The attachment method shall not weaker the frame in any manner. The top corners of the frame may be mitered and joined by spline or dowels or the four corners may be joined by open or blind mortise and tenon. All exposed edges shall be rounded a minimum of 1/8-inch The frame shall be equipped with hangers for wall hanging The hangers shall be hook and eyelet type or suitable design which will provide adequate support for each size of blackboard Not less than two hangers shall be furnished with each blackboard. The hangers shall be located not more than 6 inches from each end on the top back surface of the frame.

3 3.5.1 Chalk trough. The lower horizontal rail of the frame shall have a chalk trough. The trough shall be constructed of the wood specified in 3.2 1. The trough shall be an integral part of the frame or shall be separately constructed and securely attached to the frame and chalk board The attachment shall be by uniformly spaced flat head wood screws or gimlet point thread forming screws. Alternatively, the attachment may be by screw at each end of the trough and deformed shank nails or heavy staples spaced on 12-inch centers between the screws. The trough shall be of sufficient size to hold chalk and standard size erasers, and shall be the full length of the blackboard frame. The inside of the trough shall be of such size and configuration that (a) the chalk shall be easily removed from the trough, and (b) when an eraser is placed in the trough in a level position, directly over a piece of chalk of the size produced for general use, the bottom of the eraser will not touch the top of the chalk The exposed edges of the trough shall have a radius or chamfer of not less than 1/16-inch or more than 1/8-inch One end of the trough shall be open for cleaning, the other end may be open or closed

3 3.6 Aluminum standard and frame.

3 3.6 1 Standard The standard shall be constructed of anodized, extruded aluminum Each upright shall be supported by a base The uprights and base shall be a minimum 1-1/2 x 2 x 1/8-inch thick. Each vertical upright shall be attached to the base by two aluminum braces to prevent sway etc., and give sturdiness to uprights braces shall be a minimum 1-1/2 x 2 x 1/8-inch thick One brace on each side of the upright shall be attached to the upright and base by sheet metal screws or alternatively may be key-hole-slotted for fastening The base section of each upright shall be heliarc-welded inside the base cross-tee and heat-treated after welding or attached with four sheet metal screws There shall be a horizontal brace a minimum 1-1/2 x 2 x 1/8-inch thick, incorporating a chalk trough. connecting the two vertical uprights The bottom edge of the blackboard frame shall be not more than 4 inches above the chalk trough The horizontal brace shall provide a chalk trough accessible from both sides of the blackboard and shall run the full width. The trough shall be 2-1/4 inches in width and a minimum of 1/4-inch in depth, exclusive of the ribs or flutes. The trough shall have inside ribs or flutes to allow the accumulation of chalk dust below the level at which the eraser rests. The horizontal brace shall be securely attached to vertical uprights by means of sheet metal screws. Support shall be provided between each vertical upright and horizontal brace by means of two anodized aluminum braces, 12 inches long, 1/8-inch thick Each brace shall be attached to the horizontal brace and the vertical uprights by two No. 10 sheet metal screws. All aluminum used in the construction of the standard shall be satin finished Each opening of the aluminum tubing shall be closed with a plastic insert Alternatively, the vertical uprights may extend the full height of the chalkboard

3 3 6.2 Aluminum frame - stand-mounted blackboards The aluminum frame shall be in accordance with 3 3.6 3 except that hangers shall not be provided

3 3.6 3 Aluminum frame - wall-mounted blackboard. The blackboard frame shall be constructed of anodized extruded aluminum stock not less than 1/16-inch in thickness The aluminum shall have a satin finish The frame shall be of continuous piece totally enclosing the blackboard writing surface with the two ends securely miter-joined Alternatively, the two sides and top may be formed from one continuous piece or by three individual pieces with the bottom piece of the frame having an integral chalk trough The joining of the individual pieces shall be either mitered or butt-joined The frame shall be a minimum of 1-inch wide by 3/8-inch thick The frame shall be formed for strength and rigidity and shall be securely attached to the board to prevent movement. All joints shall be reinforced and securely joined and shall be even, smooth, and without gaps or overlap. The frame shall be equipped for wall hanging. The hangers shall be hook and eyelet type or other suitable design which will provide adequate support for the blackboard. Not less than 2 hangers shall be furnished with each item. The hangers shall be located not more than 6 inches from each end on the top rear surface of the frame. All exposed edges of the frame shall be smooth.

3.3.6.4 Aluminum chalk trough. The trough shall be constructed of the same stock specified in 3.3.6.3, not less than 1/16-inch thick with a satin finish. The lower rail of the frame shall have a chalk trough. The trough may be an integral part of bottom member of the frame or it may be of separate construction. In either case, attachment shall be by a minimum of five smooth-head post and screws evenly spaced. The trough shall be not less than 2-1/4 inches nor more than 2-1/2-inches in width and shall be the full length of the blackboard writing surface. The trough shall have a ribbed or fluted bottom along the entire length. The outside lip of the trough shall be not less than 3/8-inch in height measured from the highest

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3.4 Finish wood surface. All exposed wood surfaces shall be sanded smooth and coated with a clear lacquer or varnish. The lacquer or varnish shall be so applied as to result in a uniform glossy surface overall.

3.5 Finish of fastening devices and bolts, nuts and screws. The bolts, nuts, screws, clips and other devices for which finish is not otherwise specified herein shall be made to resist rust by the standard commercial practices of the manufacturers.

3.6 Workmanship. A high degree of craftmanship shall be exercised while performing the various operations of manufacture to eliminate the possibility of personal injury due to rough or sharp edges; to assure that the occurrence of defects will not exceed the acceptable quality level (AQL) specified herein; and to produce a finished commodity that is satisfactory in function, serviceability and appearance.

3.7 Identification marking. Identification marking shall be located on the underside of a surface. The marking shall be permanent and shall show in a plain and legible manner, the Federal stock number and the contractor's name or trademark.

3.8 Military marking. When specified (see 6.2) marking for military procurement shall be in accordance with the equipment marking requirements of MIL-STD-130 except that markings shall consist only of the letters "US" not less than 3/4-inch high and the manufacturer's identification mark not less than 3/8-inch high. Except for the manufacturer's identification mark, all marking shall be on the under surface of the chalk and eraser trough.

3.9 Instructions. The manufacturer shall furnish assembly instructions with the type I and type II, style A and style B blackboards.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated.

4.2.1 Component and material inspection. In accordance with 4.1, the supplier is responsible for insuring that components and materials used are manufactured, tested and inspected in accordance with requirements of referenced subsidiary specifications and standards to the extent specified, or if none, in accordance with this specification.

4.2.1.1 Component testing. Testing shall be performed on samples selected from each lot of each different wood components for the characteristics listed in table I. The lot size shall be expressed in units of board feet.

TABLE I. Component testing

Characteristic	Requirement paragraph	Test Method	Inspection Level	Acceptance Criteria
Moisture content	3.2.1	MIL-STD-1363	S-1	No failure permitted
Slope of grain	3.2.1	4.5.1	S-1	No failure permitted

4.2.2 End item inspection.

4.2.2.1 Visual examination. Examination of the blackboards shall be in accordance with the classification of defects and acceptable quality levels (AQL's) set forth below. All of the same type and size blackboards offered for delivery at one time shall be considered a lot for the purpose of determining the sample size in accordance with MIL-STD-105. A sample unit for this examination shall be one completely fabricated blackboard.

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TABLE II. Classification of defects

Examine	Defects	Classification Major	Minor
Wood	Species of wood not as specified.	X	
	Wood chipped or splintered	X	
	Check, crack or split over 1 inch in length	X	
	Check, crack or split up to and including 1 inch in length		X
	Any loose knot, unsound knot or knot hole.	X	
	Knots exceeding 1/4-inch in greatest dimension and closer together than 5 inches.	X	
	Excessive stain or discoloration.		X
	More than two wormholes per component, more than 1/16-inch in diameter affecting serviceability.	X	
	Loosen grain or brashness.	X	
Aluminum	Not as specified.	X	
Design	Writing surface texture does not permit ease in writing, take chalk evenly or erase easily.	X	
Workmanship and Construction	Component warped out of shape affecting appearance.		X
	Any part missing.	X	
	Malformed parts affecting serviceability.	X	
	Malformed parts affecting appearance.		X
	Broken or cracked parts	X	
	Rough or sharp edges.		X
	Pivoting device pin not aligned with hole in frame.	X	
	Locking device pin not aligned with hole in frame.	X	
	Component parts not of material specified	X	
	Any loose or unsound joints.	X	
	Joints uneven affecting appearance.		X
	Porcelain ceramic covered steel not firmly bonded to hardboard.	X	
	Composition coating not firmly bonded to hardboard.	X	
	Blackboard not securely attached to frame.	X	
	Frame does not pivot on its horizontal axis full 360 degrees	X	

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TABLE II. Classification of defects (cont'd)

Examine	Defects	Classification Major	Minor
Workmanship and Construction	Not furnished with means for locking securely in vertical position.	X	
	Not full writing surface on both sides of board.	X	
	Hangers or eyelets missing or insufficient amount.		X
Wood finish	Not coated with clear lacquer or varnish.	X	
	Not uniform or glossy surface overall.		X
	Foreign matter or grit embedded in finish.		X
	Hard runs or sags over 1 inch long on exposed surfaces.		X
	Poor adhesion, or crazing.	X	
	Area of thin film or no film.	X	
	Touch-up not neat, finish scratched, or marred on face surface.		X
	Scratch, mar, dent, machine gouge on exposed surface affecting appearance		X
	Wave resulting from uneven sanding or exposed wood affecting appearance.		X
Identification Marking	Marking missing, not permanent, illegible, incorrect, incomplete or misplaced.		X

4.2.2.2 Dimensional examination. Inspection will be made of blackboards for compliance with overall dimensions specified. Any dimension not within tolerance specified shall be classified as a defect.

4.2.2.3 Inspection levels and AQL's The acceptable quality levels (AQL's) expressed in defects per hundred units, and inspection levels shall be as follows.

	Inspection level	AQL's Major	Totals
For examination in 4.2.2.1	I	4.0	10.0
For examination in 4.2.2.2 (1 class)	I	-	4.0

4.3 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the packaging, packing and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with table III. For examination of interior packaging, the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed in table III shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot The inspection level shall be S-2 with an acceptable quality level (AQL) of 4.0 defects per hundred units.

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TABLE III. Classification of defects

Examine	Defect
Markings	Omitted, incorrect; illegible; improper size, location; sequence, or method of application.
Materials	Any nonconforming components, component missing, damaged or otherwise defective affecting serviceability.
Workmanship	Inadequate securing of contents and padding of perimeter edges. Incomplete closure of container flaps, loose strapping, distortion of container.

4.4 Testing of end items Tests shall be regularly performed as described herein on each item of blackboards being produced under the contract to assure continuous quality production. Tests shall be repeated whenever a change is made in material, design, or method of production. Additional tests shall be performed at the discretion of the Government (see 4.1).

4.5 Tests Tests shall be conducted as specified herein. Samples shall be selected in accordance with level S-1 of MIL-STD-105. Paragraph 4.5.1 specifies test requirements but does not preclude the use of other test fixtures if the mechanics of the systems are duplicated for similar tests. Certified evidence of compliance with all applicable tests must be submitted upon request. Items used in the tests must be available for examination by the inspector.

4.5.1 Slope of wood grain. The slope of grain shall be determined by combining the grain direction of two adjacent faces of a square or rectangular piece. When one of these faces is straight grained, the true or combined slope is greater than the slope on either face. Slope of grain is shown on edge grained faces by the summer wood bands, by the direction in which a free flowing ink or dye spreads, by the course taken by a narrow strip listed by a knife point and torn out. Direction of fiber may also be measured by the Teco slope grain detector supplied by the Timber Engineering Company, Inc., 5530 Wisconsin Avenue, Washington, DC 20015.

4.5.2 Gloss from wearability. The test equipment shall consist of a Taber abraser, or similar device, with the scrubbing head fitted with a standard wool felt eraser (1-1/4 X 1-1/4 inches). Chalk dust shall be rubbed into the eraser at the beginning of the test. The specimen to be tested shall be 6 inches in diameter with a 3/8-inch center hole. The specimen shall be lightly covered with chalk dust at the beginning of the test. Sixty degree specular reflectance shall be measured before the test is started. The specimen shall be subjected to 7,500 cycles on the Taber abraser. After each one-half hour of test the eraser shall be refilled with dust by hand working. After 7,500 cycles the specimen shall be removed and the surface shall be washed with detergent and hot running water to remove all imbedded material The 60-degree specular reflectance shall be measured. If the gloss has increased by more than three units, the specimen shall have failed the test.

5. PREPARATION FOR DELIVERY

5.1 Packaging Packaging shall be level A, B or C as specified (see 6.2).

5.1.1 Level A

5.1.1.1 Unit packaging.

5.1.1.1.1 Type I or Type II, Style A or Style B

5.1.1.1.1.1 Disassembly. The blackboard, vertical uprights, horizontal brace and braces shall be disassembled from the unit. At supplier's option, the complete unit may be packaged assembled.

5.1.1.1.1.2 Magnetic holders, hardware and assembly instructions. The magnetic holders, when applicable, hardware and assembly instructions shall be individually preserved in accordance with method IC-1 of MIL-P-116.

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5.1.1.1.1.3 Blackboards. One complete blackboard shall be packaged flat in a close-fitting box conforming to PPP-B-591, class II or PPP-B-636, V3c, FPF. The dimensional limitations of the folder shall be waived and the flutes may run parallel to the score line of the folder opening. A fiberboard separator pad with the fiberboard meeting the requirements of PPP-F-320 of the same length and width dimensions as the blackboard shall be utilized on top of the blackboard. Die-cut or slotted fiberboard forms with the fiberboard meeting the requirements of PPP-F-320 or interior wood blocking shall be utilized as cushioning and blocking. The disassembled components immobilized with fiberboard inserts or cells, hardware, assembly instructions and magnetic holders, when applicable, shall be secured to the separator pad. The box shall be closed, waterproofed-sealed with tape in accordance with the appendix to the box specification.

5.1.1.2 Type II, Style C or D. Each complete blackboard shall be packaged flat in a close-fitting box conforming to PPP-B-636, V3c, FPF. A fiberboard separator pad meeting the requirements of PPP-F-320 of the same length and width dimensions as the blackboard shall be utilized on top of the blackboard Die-cut or slotted fiberboard forms with the fiberboard meeting the requirements of PPP-F-320 or interior wood blocking shall be utilized as cushioning and blocking. The box shall be closed, waterproofed-sealed with tape in accordance with the appendix to the box specification.

5.1.2 Level B.

5.1.2.1 Unit packaging.

5.1.2.1.1 Type I or Type II, Style A or B.

5.1.2.1.1.1 Disassembly. The blackboard, vertical uprights, horizontal brace and braces shall be disassembled from the unit. At supplier's option, the complete unit may be packaged assembled.

5.1.2.1.1.2 Magnetic holders, hardware and assembly instructions. The magnetic holders, wher applicable, hardware and assembly instructions shall be individually packaged in a kraft paper bag and secured with tape.

5.1.2.1.1.3 Blackboards Each complete blackboard shall be wrappec in paper conforming to UU-P-268, grade B minimum basis weight 60 pounds and secured with PPP-T-76 tape The disassembled components shall be individually wrappec in paper conforming to PPP-P-291, type III and secured with PPP-T-76 tape

5.1.2.1.2 Type II Style C or D Each complete blackboard shall be wrapped with paper conforming to UU-P-268, grade b minimum basis weight 60 pounds and secured with PPP-T-76 tape

5.1.3 Level C. The blackboards shall be packaged in accordance with supplier's commercial practice

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Five unit packages of blackboards in sizes up to and including 48 by 72 inches or four unit packages in size 48 by 120 inches, shall be packed in a crate conforming to PPP-C-650, type V, style B or in a box conforming to PPP-B-601, overseas type, or PPP-B-621, class 2. A shroud shall be provided for crates, constructed as specified in the appendix to the crate specification. Each box shall be provided with a case liner constructed of material conforming to MIL-L-10547. The box or crate shall be closed and strapped in accordance with the appendix to the applicable container specification

5.2.2 Level B Each unit package of blackboards shall be packed flat in a close-fitting box conforming to PPP-B-591, class 1 or PPP-B-636, class domestic, FPF. Die-cut or slotted fiberboard forms with the fiberboard meeting the requirements of PPP-F-320 or interior wood blocking shall be utilized to protect the blackboard and chalk holder. The box shall be closed in accordance with the appendix to the box specification.

5.2.3 Level C The blackboards shall be packed to insure carrier acceptance and safe delivery at destination in containers complying with the rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special markings required by the contract or order, all containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to any special markings required by the contract or order, all containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The blackboards, covered by this specification, are intended for use in lecture, class, and conference rooms. The type I blackboard provides for ready attachment of displays, visual aids, charts or maps, as well as writing area.

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6.2 Ordering data. Purchasers shall select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, style and size required (see 1.2.1).
(c) When preproduction sample is waived (see 3.1).
(d) Color required for type II, style A or B (see 3.3.1.2).
(e) Whether casters or glides are desired for type II, style A or B (see 3.3.1.2).
(f) Whether military markings are required (see 3.8).
(g) Levels of packaging, packing, and marking required (see 5.1, 5.2, and 5.3).

MILITARY INTEREST.

Military coordinating activity:

Army - GL

Preparing activity:

GSA-FSS

Orders for this publication are to be placed with the General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 30 cents each.

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